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Mission Veridian: Gravity-Assist Trajectory Design

in a Fictional Exoplanetary System

Flight and Space Mechanics Course Assignment/Competition

Second-Year Aerospace Engineering MDM

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Course Code: R5ME2206T

Maximum Marks: 100

Prologue: The Distant Shores of Veridian

> *"The Veridian system represents one of the most intriguing and complex exoplanetary systems in our stellar neighborhood. Decades of remote sensing and astronomical observations from Earth and its vicinity have only deepened our curiosity, revealing a diverse collection of worlds unlike anything in our own solar system. Among these are multiple planets located within the habitable zone, as well as other planets that do not have solar system analogues."* [1]

Thirty light-years from Earth, the star Veridian hosts a planetary family of extraordinary scientific promise. Unlike our own solar system's orderly progression of rocky inner worlds and gas giants beyond, Veridian's architecture defies easy classification. A scorched world orbits mere millions of kilometers from the star's searing photosphere. An Earth-like planet traces a serene path through the habitable zone, its surface kept temperate by the star's steady glow. Further out, a colossal gas giant dominates the system's middle reaches, its gravity well so deep that it has shaped the orbits of everything around it for billions of years. And at the frozen edge of the system, an ice giant awaits—a pristine world of primordial ices and enigmatic moons, its secrets preserved in cryogenic suspension.

The scientific imperative to explore this system is overwhelming. The inner world offers a laboratory for extreme planetary physics. The Earth-like planet could hold the key to understanding habitable environments beyond our solar system. The gas giant's intricate moon system may harbour subsurface oceans, while the ice giant represents an entire class of exoplanets we have only begun to characterise [5].

But there is a problem.

The interstellar probe sent to explore Veridian—launched decades ago by an earlier generation of engineers—is now decelerating into the system. Its high-speed arrival, necessary to make the journey in a human lifetime, leaves it with virtually no propellant for the tour that follows [1]. Like the Voyager spacecraft before it, this explorer must rely on the gravity of the very worlds it seeks to study, bending its trajectory through carefully choreographed planetary encounters to reach its ultimate destination.

Your task is to choreograph this celestial ballet.

1 The Veridian System

1.1 The Star: Veridian

Veridian is a main-sequence star slightly more massive and luminous than our Sun. Its properties are:

Parameter	Value	Units
Stellar mass M_*	$1.05\ M_\odot$	kg
Gravitational parameter μ_*	1.393×10^{11}	$\mathrm{km}^3/\mathrm{s}^2$
Stellar radius R_*	7.0×10^5	km
Luminosity L_*	$1.2\,L_\odot$	—

1.2 The Planets

Four planets have been confirmed in the Veridian system through radial velocity and transit observations [5, 8]. Their orbital parameters are summarised below. All orbits are assumed coplanar (inclination $i = 0$) and are given in a heliocentric reference frame with the reference

direction (x-axis) aligned with a pre-alligned direction. All the ephemeris data in the ephemeris file are with respect to this heliocentric coordinate reference frame.

Table 1: Orbital and physical parameters of the Veridian system planets. Masses for Aether and Caelus are given in Earth masses ($M_\oplus = 5.972 \times 10^{24}$ kg); masses for Ventus and Glacia are given in Jupiter masses ($M_J = 1.898 \times 10^{27}$ kg).

Planet	**Semi-major Axis** (AU)	**Eccentricity**	**Mass**	**Radius (km)**	**Orbital Period** (days)
Aether	0.32	0.02	0.15 $M_\oplus$	8 500	64.5
Caelus	0.87	0.03	0.85 $M_\oplus$	7 200	289.3
Ventus	1.64	0.08	1.00 M_J	65 000	748.8
Glacia	2.75	0.06	0.10 M_J	30 000	1625.9

Orbital periods are computed from Kepler's third law, $T = 2\pi\sqrt{a^3/\mu_*}$, using $\mu_* = 1.393 \times 10^{11}$ km^3 s^{-2} and 1 AU $= 1.496 \times 10^8$ km.

1.3 Planetary Characteristics

Aether: A scorched inner world, perpetually baked by its proximity to Veridian. Its surface is likely molten on the dayside, with a thin atmosphere of vaporised rock. Not a candidate for exploration, but its gravitational influence must be considered for long-term trajectory planning.

Caelus: The Earth-analogue of the Veridian system. With a radius similar to Earth's and a mass slightly below one Earth mass, Caelus possesses a substantial atmosphere and likely liquid water on its surface [5]. It serves as the starting point for your mission—the "home base" from which your spacecraft departs.

Ventus: A gas giant of approximately one Jupiter mass, Ventus dominates the dynamical environment of the inner Veridian system. Its powerful gravity well makes it an ideal candidate for gravity-assist manoeuvres. The planet possesses a complex ring system and numerous moons, some of which may be scientifically interesting in their own right.

Glacia: An ice giant in the outer reaches of the system, Glacia represents the ultimate destination. With a mass of 0.10 M_J (approximately twice the mass of Neptune), it is a super-Neptune-class world. Its composition of water, ammonia, and methane ices preserves a record of the primordial material from which the Veridian system formed [10]. Reaching Glacia with a scientifically useful payload is the mission's primary objective.

2 Mission Overview and Objectives

2.1 Spacecraft Specifications

The interstellar probe has the following characteristics upon arrival at Caelus:

- Initial mass in Caelus orbit: $m_0 = 2500$ kg
- Propulsion system specific impulse: $I_{sp} = 300$ s (bipropellant chemical)
- Maximum available ΔV: 1.5 km/s (limited by remaining propellant)
- Scientific instruments: mass spectrometer, magnetometer, imaging system

2.2 Mission Constraints

1. **Departure:** The spacecraft begins in a circular parking orbit around Caelus at an altitude of 500 km above the planetary surface (Caelus radius $R_C = 7{,}200$ km).
2. **Destination:** The spacecraft must achieve rendezvous with Glacia—i.e., match its position and velocity vector at arrival—with a tolerance of 10^4 km in position and 0.1 km/s in velocity. Rendezvous with Glacia is a prerequisite for the spacecraft to inject itself into orbit. However, the injection into Glacia's orbit is not a part of the current mission.
3. **Maximum Mission Duration:** The total time from departure to arrival shall not exceed 8 Earth years (2922 days).
4. **Flyby Constraint:** During any gravity assist at Ventus, the spacecraft must maintain a minimum altitude of 2,000 km above the cloud tops (Ventus radius $R_V = 65{,}000$ km).
5. **Thermal Constraint:** The spacecraft must never approach within 0.4 AU of Veridian to avoid thermal damage.

2.3 Objectives

1. **Baseline Calculation:** Compute the ΔV required for a direct Hohmann transfer from Caelus to Glacia. This establishes the reference against which the gravity-assist trajectory will be compared.
2. **Gravity-Assist Design:** Find a launch window and flyby geometry such that a gravity assist at Ventus reduces the total mission ΔV below the direct transfer cost.
3. **Optimisation:** For the optimal trajectory, determine:
   - Departure date from Caelus
   - Time of flight to Ventus
   - Flyby altitude (within the allowable range)
   - Time of flight from Ventus to Glacia
   - Total mission ΔV and final mass
4. **Physical Interpretation:** Explain why the gravity assist reduces the ΔV requirement in terms of energy and angular momentum exchange.

3 Theoretical Background

3.1 Gravity-Assist Mechanics

The gravity-assist manoeuvre exploits the relative motion of a planet and spacecraft to alter the spacecraft's heliocentric velocity without expending propellant [7, 4]. In the planet-centred frame, the spacecraft approaches on a hyperbolic trajectory with excess velocity $\mathbf{v}_{\infty,\text{in}}$. After passing through the planet's sphere of influence, it departs with velocity $\mathbf{v}_{\infty,\text{out}}$ of equal magnitude but rotated by an angle δ:

$$\|\mathbf{v}_{\infty,\text{out}}\| = \|\mathbf{v}_{\infty,\text{in}}\| = v_\infty$$

The turning angle δ depends on the hyperbolic excess speed and the closest approach distance r_p:

$$\sin\frac{\delta}{2} = \frac{1}{1 + \dfrac{r_p v_\infty^2}{\mu_p}}$$

where μ_p is the planet's gravitational parameter. The spacecraft's heliocentric velocity after the flyby is:

$$\mathbf{v}_{\text{out}} = \mathbf{v}_p + \mathbf{v}_{\infty,\text{out}}$$

where $\mathbf{v}_p$ is the planet's heliocentric velocity at the encounter epoch [7].

3.2 Patched-Conic Approximation

For preliminary trajectory design, the patched-conic approximation divides the mission into phases:

1. Departure from Caelus: hyperbolic escape to achieve desired v_∞
2. Heliocentric transfer to Ventus (Lambert's problem)
3. Flyby at Ventus (instantaneous rotation in planet-centred frame)
4. Heliocentric transfer to Glacia (Lambert's problem)
5. Arrival at Glacia: hyperbolic capture or direct rendezvous

This approximation, while simplified, captures the essential dynamics and is widely used in preliminary mission design [7, 10].

3.3 Lambert's Problem

Given two position vectors $\mathbf{r}_1$ and $\mathbf{r}_2$ and a time of flight Δt, Lambert's problem seeks the conic trajectory connecting them. The solution yields the initial and final velocity vectors $\mathbf{v}_1$ and $\mathbf{v}_2$. For this assignment, you will implement a Lambert solver using the universal variable formulation [3].

4 Computational Tasks

4.1 Task 1: Ephemeris Generation (Provided)

To ensure all teams work from consistent data, an ephemeris file `veridian_ephemeris.csv` is provided. This file contains heliocentric Cartesian coordinates for all four planets at 5-day intervals over a 10-year window (MJD 60000 to MJD 63650). Columns:

```
MJD, Aether_x, Aether_y, Aether_z, Aether_vx, Aether_vy, Aether_vz,
   Caelus_x, Caelus_y, Caelus_z, Caelus_vx, Caelus_vy, Caelus_vz,
   Ventus_x, Ventus_y, Ventus_z, Ventus_vx, Ventus_vy, Ventus_vz,
    Glacia_x, Glacia_y, Glacia_z, Glacia_vx, Glacia_vy, Glacia_vz
```

Units: positions in km, velocities in km/s, time in Modified Julian Date (MJD).

4.2 Task 2: Lambert Solver Implementation

Implement a Lambert solver using the universal variable formulation. Your function should have the signature:

```
[v1, v2] = lambert_solver(r1, r2, tof, mu)
```

where `r1` and `r2` are 3D position vectors at departure and arrival, `tof` is the time of flight (seconds), and `mu` is the gravitational parameter of the central body (Veridian).

Verification case: For a Hohmann transfer from Caelus to Ventus, the time of flight is half the period of the transfer ellipse. Verify that your Lambert solver returns velocities consistent with Hohmann theory.

4.3 Task 3: Gravity-Assist Model

Implement a function that computes the post-flyby heliocentric velocity given:

```
[v_out, delta] = gravity_assist(v_inf_in, v_p, r_p, mu_p)
```

where `v_inf_in` is the incoming excess velocity vector (planet-centred), `v_p` is the planet's heliocentric velocity at encounter, `r_p` is the closest approach distance, and `mu_p` is the planet's gravitational parameter.

The function should:

1. Compute $v_\infty = \|\mathbf{v}_{\infty,\text{in}}\|$
2. Compute the turning angle δ using the formula above
3. Determine the rotation plane (defined by $\mathbf{v}_{\infty,\text{in}}$ and the planet's heliocentric position)
4. Rotate the incoming vector by δ to obtain $\mathbf{v}_{\infty,\text{out}}$
5. Add the planet's velocity to obtain $\mathbf{v}_{\text{out}}$

4.4 Task 4: Search Over Launch Windows

The relative geometry of Caelus, Ventus, and Glacia changes continuously as the planets orbit Veridian. You must search for opportunities where a Ventus flyby reduces total ΔV.

Search parameters:

- Departure dates from Caelus: every 10 days over a 3-year window (MJD 60000 to MJD 61095)
- Time of flight to Ventus: 200 to 800 days, step 10 days
- Time of flight from Ventus to Glacia: 200 to 800 days, step 10 days
- Flyby altitude: 2000 to 20 000 km above Ventus surface, step 1000 km

For each combination:

1. Use Lambert to compute the Caelus→Ventus transfer
2. Compute the incoming v_∞ at Ventus
3. Apply the gravity-assist model for the given altitude
4. Use Lambert to compute the Ventus→Glacia transfer from the post-flyby state
5. Sum the ΔV for departure from Caelus, any deep-space manoeuvres (if needed), and capture at Glacia
6. Record the total ΔV and mission duration

4.5 Task 5: ΔV Calculations

Departure from Caelus: From circular orbit at radius $r_C = R_C + 500$ km to hyperbolic excess velocity $v_{\infty,\text{dep}}$:

$$\Delta V_{\text{dep}} = \sqrt{v_{\infty,\text{dep}}^2 + \frac{2\mu_C}{r_C}} - \sqrt{\frac{\mu_C}{r_C}}$$

where $v_{\infty,\text{dep}} = \|\mathbf{v}_{\infty,\text{dep}}\|$ and μ_C is Caelus's gravitational parameter.

Arrival at Glacia: From hyperbolic approach with excess velocity $v_{\infty,\text{arr}}$ to circular orbit at radius $r_G = R_G + 500$ km $= 30{,}500$ km:

$$\Delta V_{\text{arr}} = \sqrt{v_{\infty,\text{arr}}^2 + \frac{2\mu_G}{r_G}} - \sqrt{\frac{\mu_G}{r_G}}$$

Deep-Space Manoeuvres: If the Lambert solution requires a mid-course correction, include it as additional ΔV.

5 Data and Constants

Table 2: Physical constants for the Veridian system.

Quantity	**Value**	**Units**
Veridian gravitational parameter μ_*	1.393×10^{11}	km^3/s^2
Caelus gravitational parameter μ_C	3.986×10^5	km^3/s^2
Ventus gravitational parameter μ_V	1.266×10^8	km^3/s^2
Glacia gravitational parameter μ_G	1.267×10^7	km^3/s^2
Caelus radius R_C	7,200	km
Ventus radius R_V	65,000	km
Glacia radius R_G	30,000	km
1 AU (for reference)	1.496×10^8	km
Astronomical day	86,400	s

6 Deliverables

6.1 Report (Maximum 5 Pages)

1. **Introduction:** Brief overview of the mission and your approach
2. **Methodology:** Description of your Lambert solver, gravity-assist model, and search strategy
3. **Results:**
   - Direct Hohmann transfer ΔV (baseline)
   - Optimal gravity-assist trajectory parameters (dates, TOFs, altitude)
   - Total ΔV and final mass
   - Comparison with direct transfer
4. **Physical Interpretation:** Explain why the gravity assist reduces ΔV in terms of energy and angular momentum [4, 9]
5. **Verification:** Demonstrate that your Lambert solver reproduces the Hohmann transfer test case
6. **Plots:**
   - Pork-chop plot showing total ΔV as a function of departure date and time of flight to Ventus

- Heliocentric trajectory plot showing the optimal path
- Velocity vector diagram at the Ventus flyby

6.2 Code

Submit well-commented source code (Python or MATLAB) that implements all calculations. Include a `README` file with instructions for running your code.

6.3 Data File

Submit your optimal trajectory parameters in the following format:

```
departure_mjd, tof_ventus, altitude, tof_glacia, deltaV_total
```

7 Grading Rubric

Criterion	Weight
Correctness of Lambert solver (verification)	20%
Correct implementation of gravity-assist model	20%
Successful identification of gravity-assist opportunity	25%
Quality of physical interpretation	15%
Report clarity and plots	10%
Code quality and documentation	10%

8 Hints and Suggestions

- **Start simple:** Verify your Lambert solver on the Hohmann case before attempting the full search.
- **Vectorise operations:** Use NumPy (Python) or vectorised operations (MATLAB) to speed up the search loops.
- **Coarse-to-fine search:** Begin with a coarse grid (20-day steps) to identify promising regions, then refine around the minima.
- **Flyby geometry:** The rotation direction matters. Passing behind the planet (relative to its motion) increases energy; passing in front decreases it [4]. For reaching an outer planet, you want to increase energy.
- **Documentation:** Comment your code as if someone else will need to understand it—because your grader will.

9 Software Policy

Try to develop your code, albeit using well-known python/matlab libraries and functions of astrodynamics. You might need an established function for the Lambert's solver. Using GMAT, FreeFlyer is prohibited and will not be accepted in your submission.

Epilogue: The Art of the Slingshot

> *"The gravity-assist manoeuvre is one of the most elegant techniques in space exploration—a celestial collaboration between human ingenuity and planetary motion. As your trajectory curves around Ventus, gathering speed from the giant's orbital momentum, you will witness firsthand the poetry of celestial mechanics: energy conserved, yet transferred; momentum preserved, yet shared. This is not merely an exercise in computation; it is a glimpse into the fundamental workings of the cosmos."*

The light of Veridian awaits. Navigate wisely.

February 2026

References

[1] GTOC 13 Problem Statement. NASA Jet Propulsion Laboratory, 2025. https://gtoc.jpl.net/gtoc13/

[2] Analysis of Exoplanet Systems in Primary-centric Framework. Zenodo, 2025.

[3] Spacecraft Orbital Mechanics (SESA6076) Module Description. University of Southampton, 2026.

[4] Hibberd, A. "3I/ATLAS: How low should we go?" Initiative for Interstellar Studies, 2025.

[5] Lubin, J. et al. "The HD 191939 Exoplanet System is Well Aligned and Flat." *Astronomical Journal*, 168(5), 2024.

[6] GTOC 1 Problem Description. ESA Advanced Concepts Team, 2013.

[7] Lantukh, D.V. "Preliminary design of spacecraft trajectories for missions to outer planets and small bodies." UT Austin Thesis, 2015.

[8] Lubin, J. et al. "The HD 191939 Exoplanet System is Well-Aligned and Flat." arXiv:2409.06795, 2024.

[9] "Slingshot to the Outer Planets." TeachEngineering.org, University of Colorado Boulder.

[10] Lusvarghi, M. et al. "Enceladus moon in situ science: preliminary mission analysis for the EPOPEA mission." IAC-23, 2023.